MD Redwan Hossain

redwanhossain.seu@gmail.com   +49 15753195710

Vierzigmannstr. 10, 91054, Erlangen, Germany   LinkedIn   GitHub

Google Scholar

PROFILE

A passionate and ambitious individual with expertise in data analytics, machine learning, AI, and web development. I am committed to mastering Python, SQL, statistical modeling, ML & AI. Skilled in data preprocessing, feature engineering, and model optimization using Scikit-Learn, PyTorch, and Power BI. Brings a strong mix of technical expertise in Data Science and web development background to deliver practical, data-driven business solutions.

PROFESSIONAL EXPERIENCE

Master Thesis Student 2025/11 – 2026/05 | Erlangen

Department Mathematik

My Master's thesis explores how public transport planning can become more efficient and sustainable when electric buses and conventional buses are scheduled together. The main challenge was that electric buses cannot be planned like normal diesel buses. They have limited battery range, need charging time, and depend on available charging infrastructure at terminals or depots. Because of this, timetable planning, vehicle scheduling, fleet assignment, and charging decisions should not be handled separately. In my thesis, I compared three approaches: a greedy heuristic, a hybrid MIP method, and a fully integrated charging-aware mixed-fleet MIP. The integrated model gave the strongest results because it considered routing, battery feasibility, charging, and capacity constraints together. This work allowed me to combine mathematical optimization, data science, and sustainable mobility using Python, Pyomo, Gurobi, and py5.

Web Developer 2020/03 – 2021/12 | Dhaka, Bangladesh

Aarashi Online Store

- **API Development:** Developed scalable Web APIs and tested them using Postman.
- **Database Design:** Designed and developed the database structure.
- **UI Development:** Created user interfaces integrating RESTful APIs and HttpClient.
- **Documentation:** Produced comprehensive documentation for the application.

Skills: JavaScript Framework, Responsive Design, Backend Development, API Development, Postman, SQL Server, Data Warehouse, DevOps, Cloud Service, Modern Frameworks

Software Engineer Trainee 2019/12 – 2020/02 | Dhaka, Bangladesh

Unlocklive IT

- **API Development:** Assisted in developing and testing scalable Web APIs.
- **Database Design:** Participated in designing and implementing databases.
- **UI Development:** Contributed to creating user interfaces using RESTful APIs.

Skills: JS, CSS, Laravel, PHP, MySQL

EDUCATION

M.Sc. in Data Science (Major Machine Learning & AI) 2026/05 | Erlangen, Germany

University of Erlangen-Nuremberg

Pattern Recognition | Mathematics of Learning | Pattern Analysis | Machine Learning in Signal Processing | ML for Time Series | Deep Learning | Advanced Anomaly Detection for ML | High-dimensional Statistics | Control Machine Learning & Numerics | Statistics | Business Intelligence

B.Sc. in Computer Science & Engineering 2016 – 2020 | Dhaka, Bangladesh
Southeast University, Bangladesh
Data Structure | Advanced Algorithm | C++ | Python | SQL | Java | OOP | Information & Software Development | Calculus | Numerical Methods | Data Mining | Linear Algebra | Artificial Intelligence | Computer Graphics | Advance Networking

PROJECTS

Using Machine Learning to Classify Students and Predict Student Purchases

- Developed a machine learning pipeline to predict student conversion from free to paid plans using engagement data.
- Trained and evaluated Logistic Regression, KNN, SVM, Decision Tree, Random Forest
- Applied feature engineering (VIF-based feature selection, categorical encoding, scaling) to improve model performance.
- Achieved strong predictive accuracy and ROC-AUC > 0.8, enabling targeted marketing and improved customer retention strategies

Heart-Disease-Prediction-Using-Machine-Learning

Developed a machine learning classification model to predict the presence of heart disease based on patient health metrics.

- Trained and evaluated multiple algorithms (Logistic Regression, KNN, SVM, Decision Tree, Random Forest) using accuracy, precision, recall, and ROC-AUC.
- Technologies used: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Jupyter Notebook.

Bulldozer-Price-Prediction-Using-Machine-Learning

- Built an end-to-end machine learning regression model to predict bulldozer auction prices using historical sales data.
- Trained and optimized models (Random Forest Regressor, Gradient Boosting) with cross-validation and hyperparameter tuning.
- Technologies used: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Jupyter Notebook.

PUBLICATIONS

SafeBirth AI: Tuned Explainable Ensembles for Maternal and Fetal Health Classification. 2025/12/22
IEEE WIECON-ECE, Accepted for Publication (2025) [*Not yet live*]

- **Model Development**: Designed a voting ensemble of **XGBoost, Random Forest, and Decision Trees,** achieving **98.39% accuracy**.
- **Data Engineering**: Resolved class imbalances using a hybrid **SMOTE+ROS** approach and optimized parameters via **Bayesian/GridSearch**.
- **Explainable AI**: Integrated **SHAP and LIME** to provide global and local interpretability for trustworthy clinical decision-making.
- **Performance Validation**: Conducted comparative analysis across CTGs and Maternal datasets using **stratified cross-validation** and **ROC-AUC metrics**.

Protecting Regular and Social Network Users in a Wireless Network by Detecting Rogue Access Point: Limitations and Countermeasures 2021
Taylor & Francis Group
Authored a chapter in "Securing Social Networks in Cyberspace" (CRC Press, 2021), focusing on detecting and mitigating rogue access points (RAPs) in wireless networks, including experimental results and security measures.

SKILLS

Python, C++, SQL, ReactJs, Laravel

Microsoft Office 365

Pandas, NumPy, Matplotlib, Scikit-Learn, Pytorch, Seaborn, Pytorch, PowerBI, Data Visualization, Cloud Service, Cloud Platform, Model Deployment, Big data , Data Engineering, MLOps, Business Intelligence, Data Warehouse, Statistical Analysis

LANGUAGES

English (Fluent)
IELTS 6.5

German (A2)

Bengali (Native)